Automotive System Evaluation

Lecture:

- **User Experience in Automotive Product Development**

Aachen, 10.04.2018

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Introduction - Acceptance

The increasing amount and complexity of systems, components and functions require a **user-centred optimization** as design maxim.

- Today there are already about 1200 - 2000 particular functions in a modern passenger car.
- New technical options offered in the field of driving information and assistance systems will further increase this number.
- In addition, new technologies (E-mobility, integration of mobile devices etc.) require greater interconnection with environment and infrastructure, which again results in higher complexity of functions as well.
- Such a complexity demands a optimization of every single function towards the user, as well as regarding these as parts of a **complete system** that has to be tended by the driver.
- Systems of this complexity should constitute a complete system that can be used in a convenient, efficient and safe way.

VW Beetle, ca. 1950

Infotainment
VW Golf 5

Navigation,
Driver Assistance
BMW 7er (2008)

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Introduction

- 15/10/2011
Abu Dhabi // Breakdown of BlackBerry services for 3 days
- While the messaging service was not available for BlackBerry users, accident rates in Dubai decreased by 20%.
- At the same time, accident rates in Abu Dhabi decreased by 40% and no lethal accidents were recorded.
- On average, Dubai states an accident every 3 minutes and a lethal accident every 2 days.
- Lt. Gen Dahi Khalfan Tamin, head of the police department Dubai, and Brig Ben Hussein Al Herethi, head of the police department Abu Dhabi, both saw a connection between the decreasing accident rates and the BlackBerry service breakdown between Tuesday and Thursday:
"Roads got safer when BlackBerry stopped working."

* Quelle: winfuture.de

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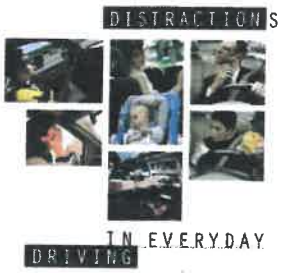
Introduction - Safety

Effectiveness and Safety from the user's perspective increasingly become a n°1 topic and a distinguishing quality for technical systems:

- The driver's actual task is driving. Besides, he is confronted with secondary tasks, like the use of additional functions while driving.
- These can lead to distraction from the driver's primary task – with accidents as result.
- Besides setting the goal of mere technical/ functional optimization of components or systems, there is also an urgent necessity to ensure security in using these.



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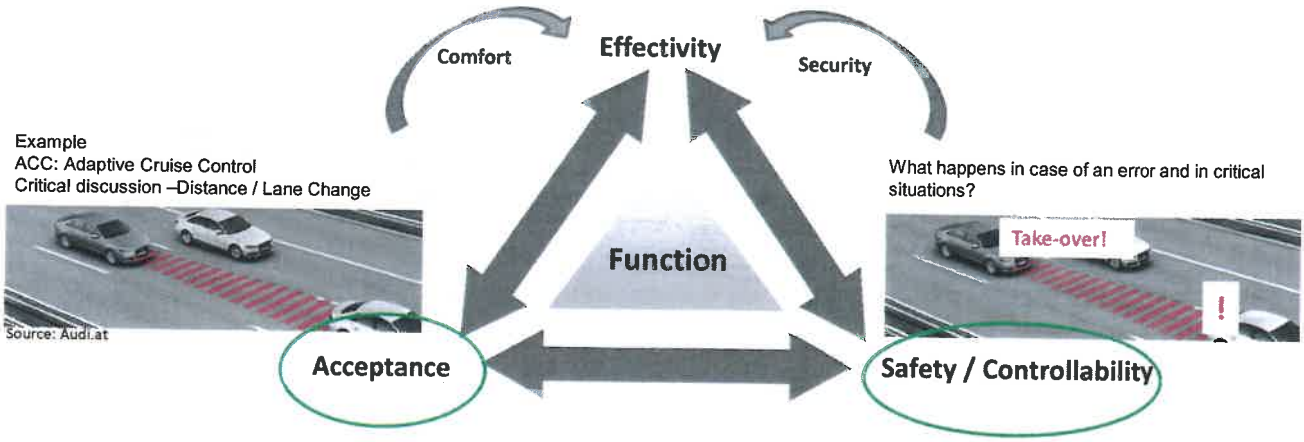
Introduction - The Challenge:
Bringing effectiveness, acceptance and controllability in line



... are to be vaildated via empirical surveys!

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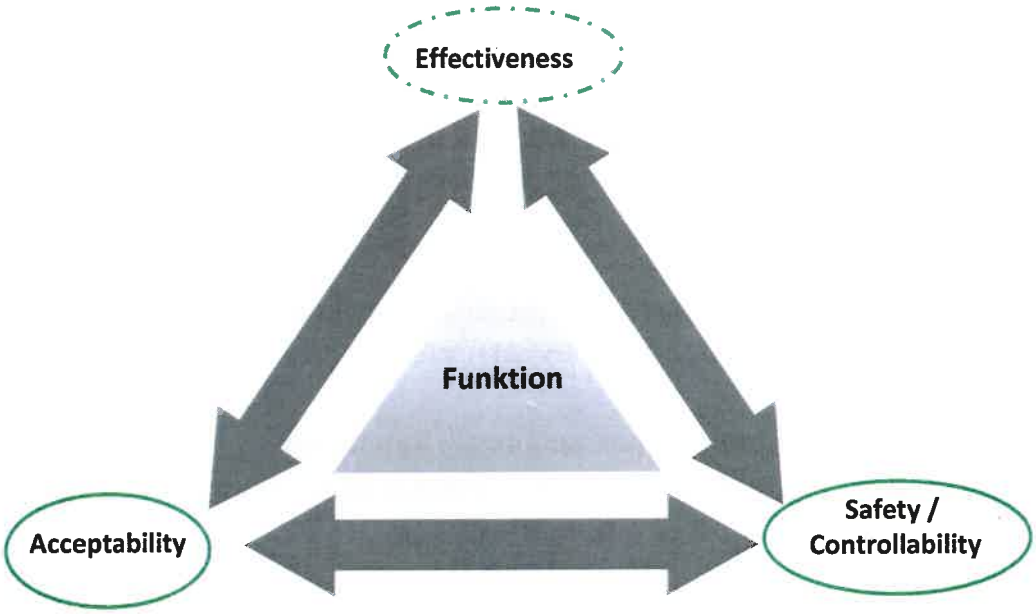
Introduction - The Challenge:
Bringing effectiveness, acceptance and controllability in line



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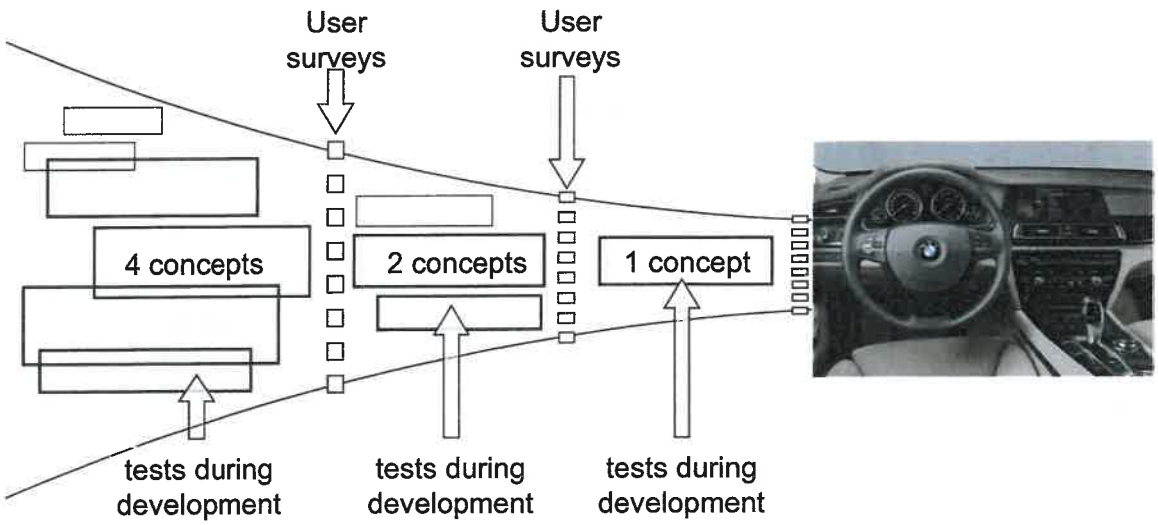
Introduction - User Evaluation in the Design Process

During the whole process of product development, it is necessary to constantly involve representative users.



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Introduction
Aspects of System Evaluation

Acceptance	= Does the user accept the system?
Safety / Controllability	= Can the system be handled by the user without risk in any situation?
Effectivity	= Does the system do what it is supposed to?
Efficiency	= How many resources are needed to fulfil a task?
Intuitivity / Learnability	= How intuitive is the usage?
Fun to use	= How is the usage of the system experienced by the user?

On the level of...

Complete Systems
(e.g. complete vehicle)

Functions
(e.g.driver assistance / driver information systems)

Components
(e.g. steering, braking, recuperation)

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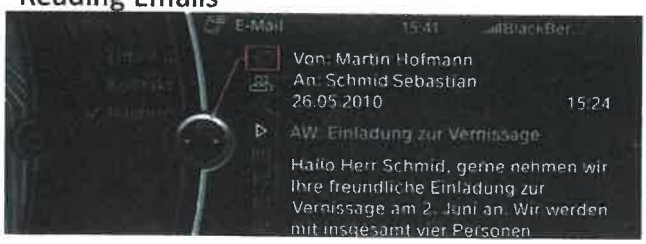
Safe to use while driving?

Interactive Handbook – BMW

TV – Function

Reading Emails



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Intuitivity

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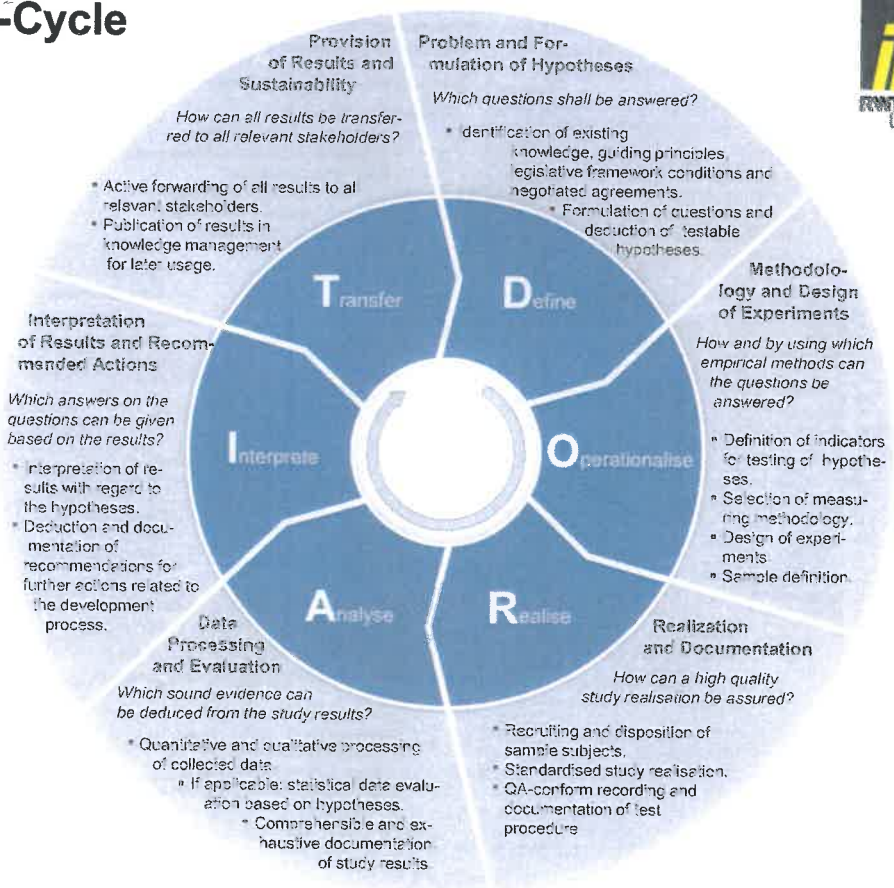
DORAIT-Cycle



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Activity !

- Each group formulates <u>one (the) most important research question</u> towards user experience for their topic (just collecting)!

  - Group 1:
  - Group 2:
  - Group 3:
  - Group 4:
  - Group 5:

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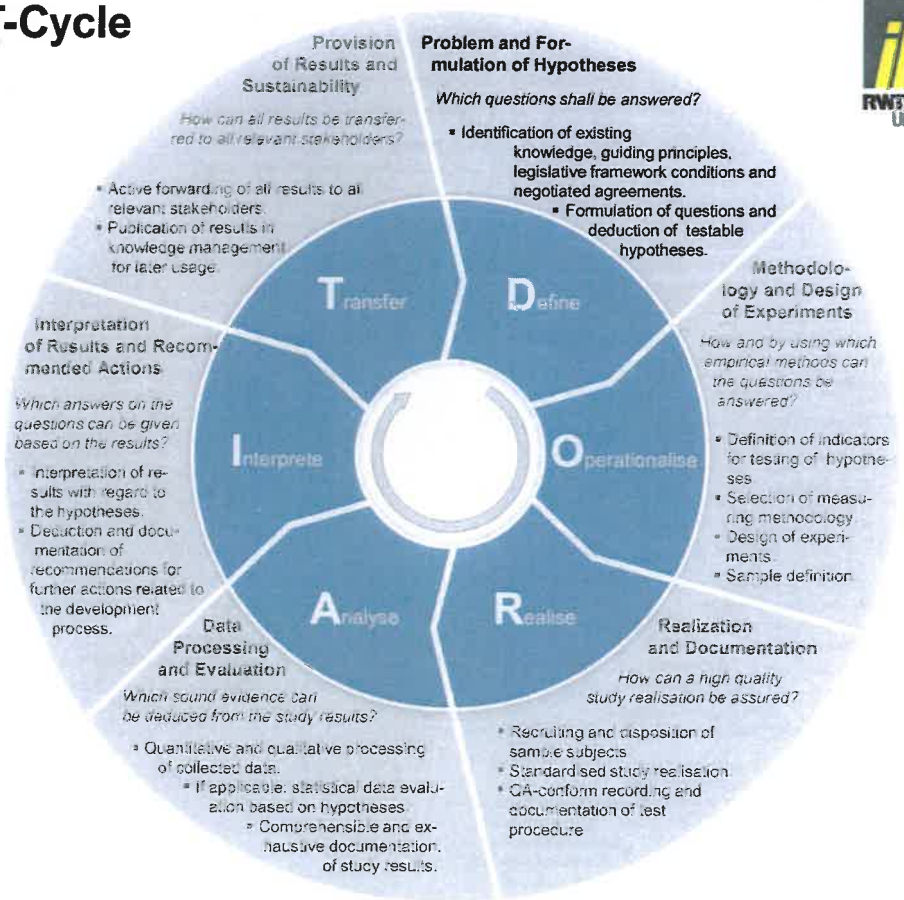
DORAIT-Cycle



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Define
An example ....

Controller

vs

Touchscreen

What would be a proper hypotheses* for a test?

*Definition Hypotheses = assumptions, guesses or unproven presuppositions that can be verified or falsified through empirical testing

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Define
Hypotheses

- Scientific hypotheses have to meet certain **criteria**. These include:

1. A scientific hypothesis refers to real facts that can be empirically examined, i.e. that can be operationalised (= the included concepts match observable events or behavior).
2. A scientific hypothesisis is a universal assumption, i.e. it exceeds single cases or singular events(≫universal sentence≪).
3. A scientific hypothesis has to be underlain at least implicitly by the formal structure of a reasonable conditional sentence (≫if-clause≪, ≫the more…, the…≪)
4. The conditional clause has to be potentially falsifiable, i.e. it must be possible to consider circumstances / outcomes with the potential to contradict the conditional clause.

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Define
Hypotheses

- According to these criteria, the following statement is a proper scientific hypotheses:
  - »Frustrated people react aggressively.«

    The respective conditional clause states: »If people are frustrated, they react aggressively.«

    The statement refers to a real, empirically examinable fact, claims universal validity and is falsifiable.

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Define
Hypotheses

- According to the criteria, the following statement is not a proper scientific hypotheses:
  - »There are children who never cry.«
    - Reformulated, it states: »There is (at least) one child that never cries.« This means that criteria 4 (falsifiability) is not met:

      The sentence could only be falsified if it was possible to show for every child in the world that they cry at some point of time. Because, realistically, this proof can never be made, the sentence can not be falsified (same goes for every other „There is…“ sentence).

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Define
Hypotheses

- According to the criteria, the following statement is not a proper scientific hypotheses:

  ≫Heavy smoking can lead to a heart attack.≪

  - This sentence (like all „can"-clauses) is not falsifiable either, because every possible event – whether a smoker gets a heart attack or not – is in accordance with the „can"-clause. The sentence is correct at all events and therefore tautological.

    If the hypothesis stated ≫If people smoke a lot, they have a higher risk of getting a heart attack than people who smoke little or not at all≪, the statement would be falsifiable.

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Define
Our example ....

Controller

vs

Touchscreen

„The use of a controller during a driving task is safer than the use of a touchscreen."

Precise definition of the concept of safety is needed
To test this hypotheses it has to be transformed into an **empirical hypothesis**

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Define
Our example ....

Controller

VS

Touchscreen

Empirical hypotheses:
„While driving *the mechanical input of alphanumeric characters via controller causes less gaze deviation from the driving task than when using a touchscreen.“*

The fit of operational control in the driving task is reflected in the measurable indicator of gaze deviation from the driving task.
This is called ***Operationalisation***

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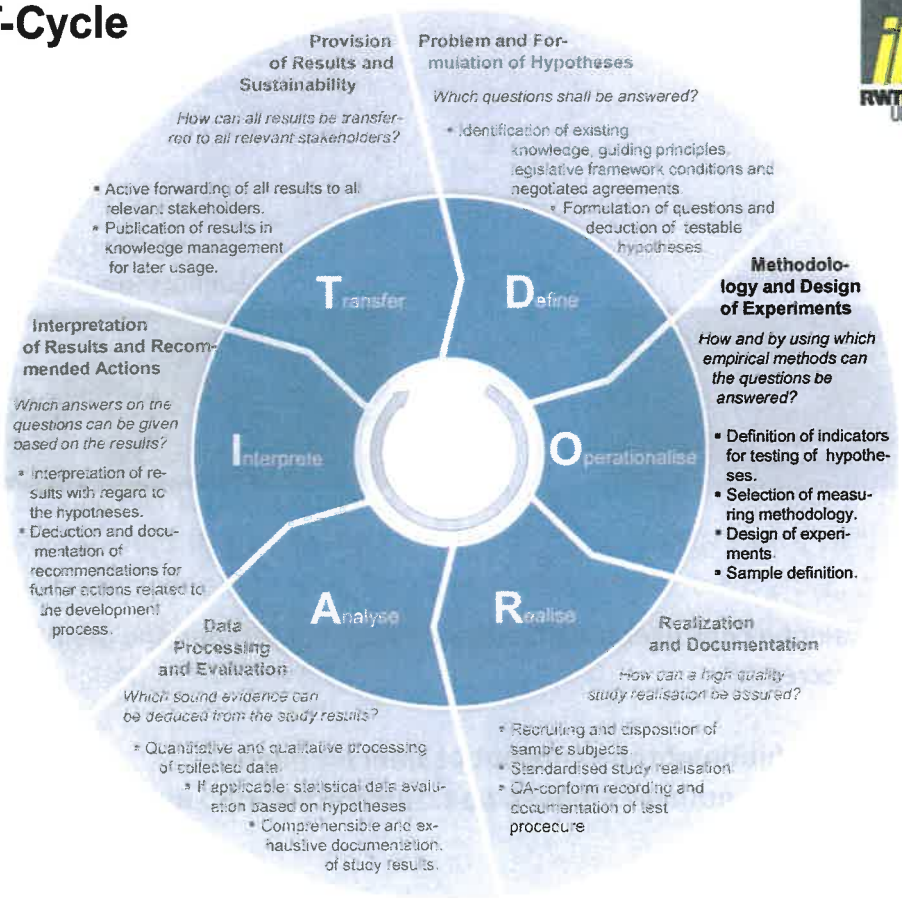
DORAIT-Cycle



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Operationalise
How to measure distraction from the driving task ...

- Gaze deviation from driving task
- Driving performance (lane deviation, lane crossings etc.)
- Subjective ratings
- Cognitive workload
- Physical workload
- etc ..

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Operationalise
... but how to measure hunger?

- Duration of food deprivation
- Percentage of blood sugar
- The amount of food consumed when given the opportunity to eat
- How fast food is consumed when given the opportunity to eat
- Measuring of stomach contraction
- What kind and how many unpleasant entities is someone willing to put up in exchange for something to eat?
- What kind of or how many pleasant entities are disdained in favour of something edible?
- Subjective ratings
- Measuring of salivation when confronted with food
- ...

→ For every empirical question it has to be examined what suitable measures exist to validly answer the question.

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Activity II

- Each group reformulates their research question towards a testable (operationalized) hypotheses!

  - Group 1:
  - Group 2:
  - Group 3:
  - Group 4:
  - Group 5:

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Operationalise
Selecting a Sample

Population

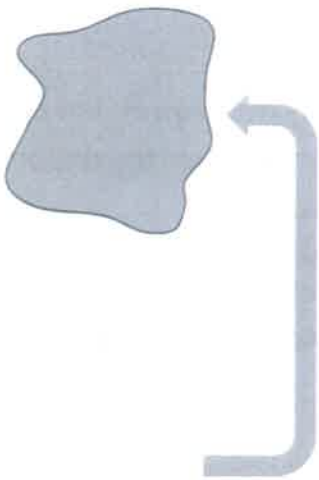
All individuals about whom a conclusion is to be drawn.

Inferential Statistics

Conclusions about the population are derived from sample.

Sample

Cutout of population that is examined.



Example

To investigate how satisfied students of RWTH University are with their studies, the students of one course are questioned.

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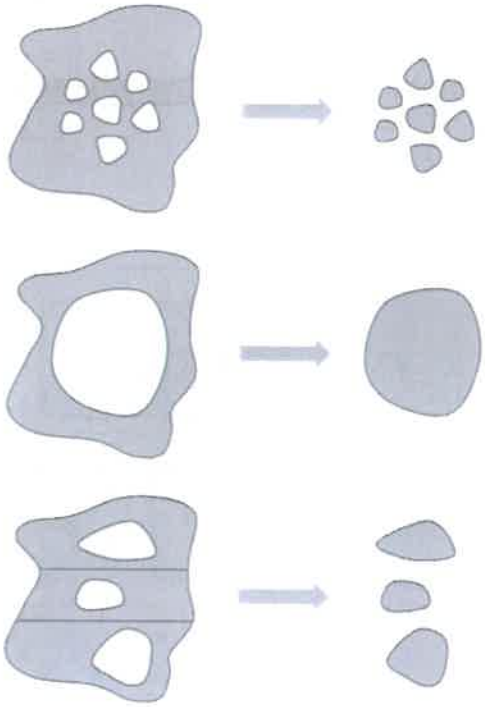
Operationalise
Selecting a Sample



Random Sample

Elements are chosen randomly:

Each element of the population has an equal chance of being selected.

Chunk Sample

Natural groups (e.g. university courses) are selected and fully examined.

Stratified random Sample

Sample **proportionally corresponds** to population with respect to certain traits.

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Operationalise
Selecting a Sample

Chunk Sample

Advantage:

low organisational efforts

Disadvantage:

sample might be selective

RWTH example: Highly frustrated students don't show up in certain courses.

Random Sample

If the selection is perfectly randomised, the sample reflects the population's distribution very well.

Problem: All elements must be known in advance.

RWTH example: Random choice of students with the aid of registrar's office lists.

Stratified random Sample

In the case of good aquisition, sample reflects population very well.

Problem: Proportions of population must be known.

RWTH example: Selection of students on the basis of university satistics.

All types of samples have pros and cons. The decision to assign subjects in a certain manner has to be made with consideration to available ressources and the desired precision.

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Activity III

- Each group defines the required sample to test their hypotheses!

  - Group 1:
  - Group 2:
  - Group 3:
  - Group 4:
  - Group 5:

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Operationalise
Methods of User-Centred Evaluation

External Validity

Reliability

(abstract) Driving Task	Field Operational Test	Evaluation inside of the vehicle, on • Public roads • Test tracks	Qualitative & quantitative feedback Objective driving performace data
	Controlled Field Test		
	Dynamic Driving Simulation	Evaluation in Driving Simulators (AAM, ESoP Richtlinien)	Qualitative & quantitative feedback Objective driving performace data
	Static Driving Simulation		
	Lane Change Task		
	Occlusion	Evaluation in laboratory context (PC, Desktop, mock-ups, drafts)	Objective driving performace data
	Eyetracking		
	Pupillometry		
	PDT (Peripheral Detection Task)		
No Driving Task	Expert Evaluation		Qualitative & quantitative feedback
	Questionnaires & inventories		
	Personal interviews		

Needed Ressources for Study

Necessary maturity level of prototype

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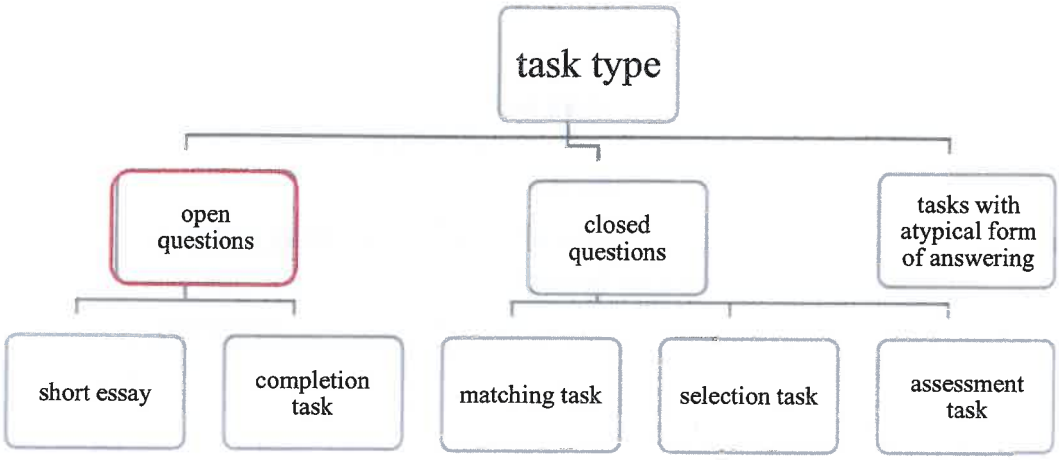
Operationalise
Questionnaires and Inventories



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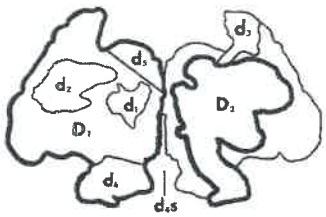
Operationalise
Open Questions

Location (L)

The following 3-point system is used for scoring Location, or the tendency to break down the inkblot into smaller fragments.

0 – use of the whole blot, or at least all except minor portions of the blot (cut-off whole)
1 – use of a large area of the blot, such as one entire side or the entire center of the blot
2 – use of smaller areas of the blot

Card No. 18A



Location	Poor (0)	Fair (1)	Good (2)
W	butterfly; Santa Claus; a house	Spanish dancers with whirling capes; relief map; rocks; face of a sad animal	
W	crab with split abdomen; man with beard	mask; 2 men fighting; table with food on top	
D_1	a fish; a telephone	rock formation; gnome carrying sack on shoulders; cotton ball (card sideways)	
D_2			side view of angel praying
d_3	dog	chicken; sea shell	
d_2		moss; aerial view of forest; mold; head of sheep	
d_1		torch	piece of cauliflower; a tree
$d_{1,5}$		oriental headdress	ornate lamp base
d_5		man's head	
d_6		rabbit peering out from behind rock; foot in toga	head of lamb (card inverted)

Holtzman Inkblot Technique as an advancement of Rorschach Tests.

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Operationalise
Open Questions

Specifics: Questions that can be openly answered do not offer alternative answers. The answer has to be formulated, that is produced, by the questioned person.

Pro: Reproduction of knowledge or generation of answers is necessary - thus, it is impossible for the questioned person to simply guess or randomly answer the question.

Contra: Interpreting the given answers takes much effort and time from participant, conductor and interpreter;

ambiguity of answers may cause as reduced objectivity in interpretation

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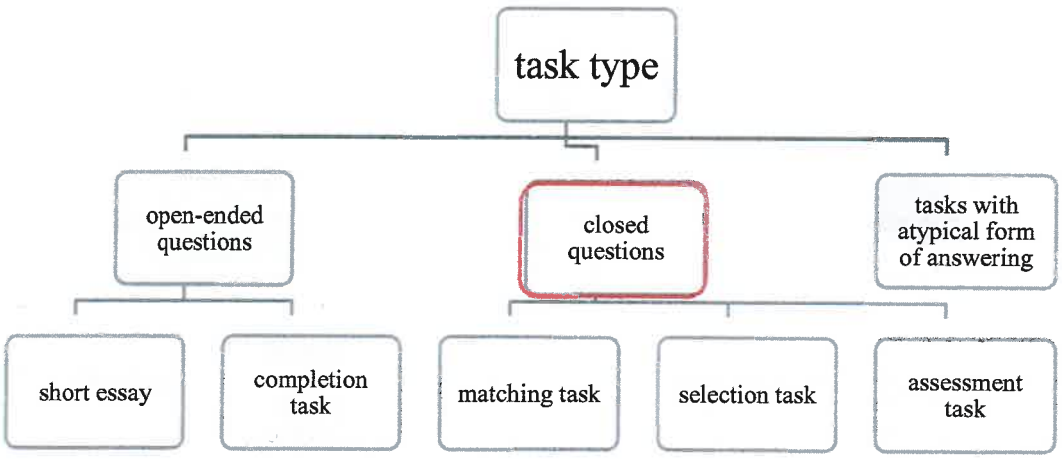
Operationalise
Questionnaires and Inventories



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Operationalise
Rating Scales

Differences in graduation

Analogue Scale

-2 +2

Rating Scale

-2	-1	0	+1	+2

Differences in polarity

Semantic Differential (bipolar)

-2	-1	0	+1	+2

Likert-Scale (unipolar)

0	1	2	3	4

Layout differences in given scale

Verbal

never	now and then	always

Numerical

-1	0	+1

Optical

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Operationalise
Closed Questions

Specifics: Closed questions provide several alternative answers.

Pro: demands low effort and time investment of subject, instructor and interpreter;

interpretation of answers is nonambiguous and therefore objective

Contra: No knowledge (re)production is needed to answer closed questions – this means that guessing and random answering on the part of the subject are possible .

Alternative answers should represent **exhaustive and disjunctive** categories, i.e. the chosen alternatives should capture all possible answering tendencies and must not overlap.

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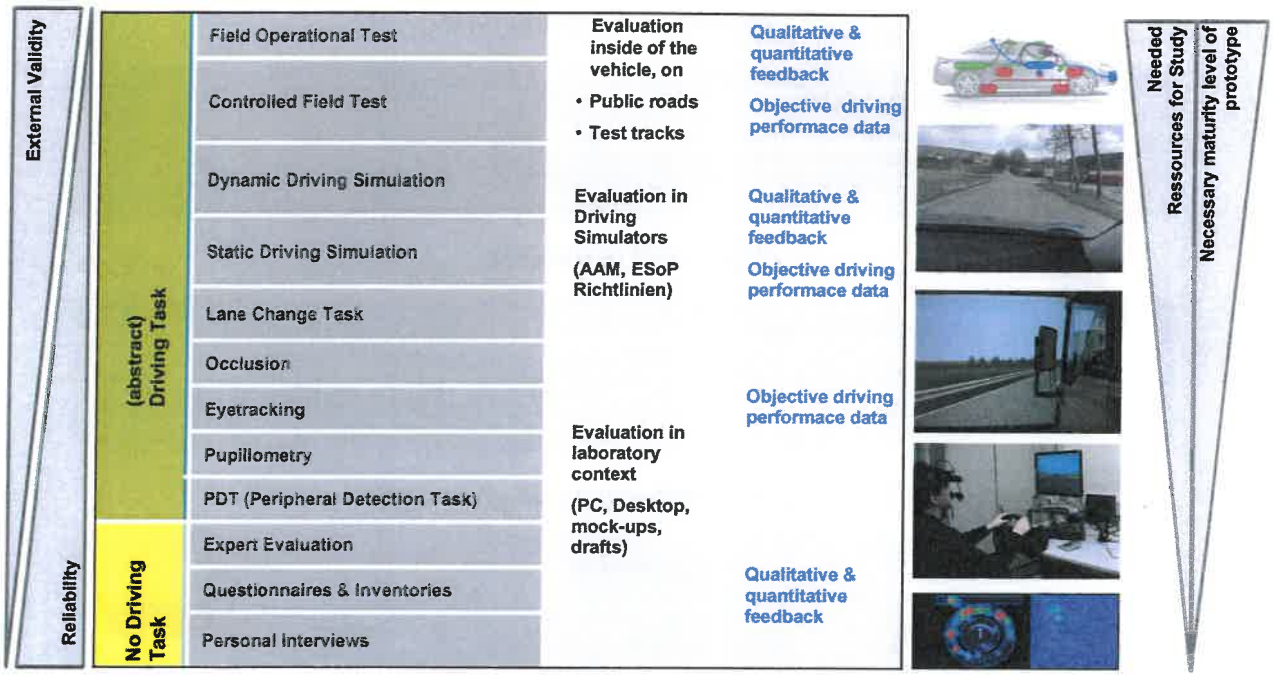
Operationalise
Methods of User-Centred Evaluation



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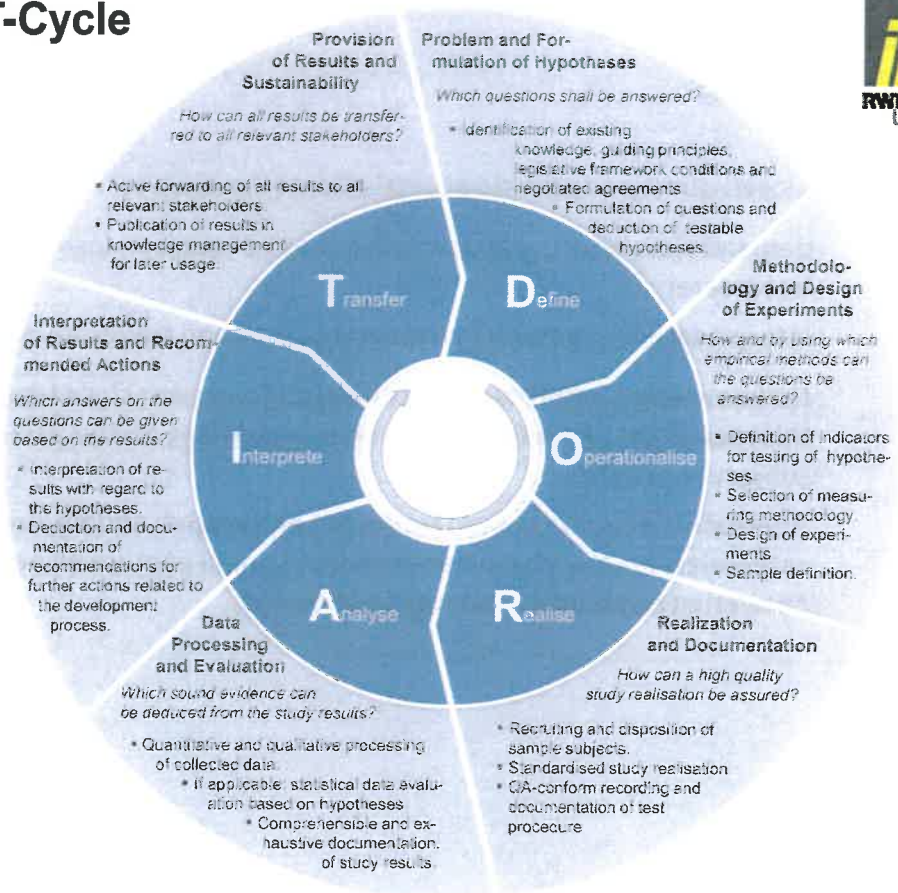

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End

Thank you for your attention!

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